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Description:

The aim of this project is to create a Parking Spot Booking Application which helps users search and reserve parking spaces. It is a multi-user app. There can be multiple Users but only one Admin. Admin and Users have their own profile pages.

Users can search for available parking spots based on location or pincode and can book or release parking spots for four-wheel vehicles only. Users can also view their active bookings and complete booking history from their dashboard. A personalized summary page allows users to view the count of their currently occupied and previously closed bookings.

Admin has full control over the management of parking venues and individual parking spots.

Admin can create, update, or delete venues, and manage parking spots within each venue. The system displays the latest added parking spots along with their availability status. A search functionality is included, allowing users to search by location, pincode, or venue name.

Technologies used:

Frontend – HTML and Bootstrap have been used for styling and responsiveness of app.

Template Engine - Jinja2 has been used for HTML template generation.

Backend - The application uses Flask, SQL Alchemy to implement core functionalities, in addition.

Data Storage - The application uses SQLite for data storage.

Architecture and Features:

The project has an application folder in root which contains a templates folder which has all the templates, a directory folder which contains the database file (SQLite) of Parking Booking Application, an application folder which has python files of controllers, and a document folder which contains project report Further a app.py file to run the application.

All **CRUD** operations are fully implemented. Admin can manage venues and parking spots.

Users can book and release spots for four-wheelers, track their active and past bookings, and view booking summaries. The admin has access to a revenue summary page showing income from each venue.

DB Schema Design:

1. Users table

Column Name	Type	Constraints
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id	Integer	Primary key, Auto Increment
email	String	Unique, Not Null
password	String	Not Null
name	String	Not Null
address	String	Not Null
pincode	Integer	Not Null
role	Boolean	Not Null, Default = 1

2. Parking Lot table:

Column Name	Type	Constraints
id	Integer	Primary key
location	String	Unique, Not Null
price	Integer	Not Null
pincode	Integer	Not Null
available_spots	Integer	Not Null

3. Parking Spot table:

Column Name	Type	Constraints
id	Integer	Primary key
lot_id	Integer	Foreign key, Not Null
status	String	Not Null

4. Reserve Parking Spot table:

Column name	Type	Constraints
id	Integer	Primary key
spot_id	Integer	Foreign key, not Null
user_id	Integer	Foreign key, not Null
vehicle_number	String	Not Null
parking_timestamp	Datetime	Not Null
leaving_timestamp	Datetime	Null
parking_cost	Integer	Null

[Video Link](#)